PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Is Short-term and Long-term Exposure to Black Carbon Associated with Cardiovascular and Respiratory Diseases? A Systematic Review and Meta-Analysis based on Evidence Reliability
AUTHORS	Song, Xuping; Hu, Yue; Ma, Yan; Jiang, Liangzhen; Wang, Xinyi; Shi, Anchen; Zhao, Junxian; Liu, Yunxu; Liu, Yafei; Tang, Jing; Li, Xiayang; Zhang, Xiaoling; Guo, Yong; Wang, Shigong

VERSION 1 – REVIEW

REVIEWER	Hualiang, Lin Sun Yat-sen University
REVIEW RETURNED	16-Mar-2021

GENERAL COMMENTS	This is an interesting topic to summarize the health impact the PM component, particularly the EC component. A few language issues should be noted, for example: "short-term exposure to BC or EC were related" were - was The authors mentioned tha PM2.5-adjusted and unadjusted results, I wonder why PM2.5 mass cocentration should be adjusted in examining the component of PM2.5, can the authors comment on thiis. It is interesting to observe that the effects were larger among Asian population, why stronger effects were detected in Asia, more explanations are needed. A few systematic reviews have been available in the literature, I agree that it is important to update the evidence, the authors, could please, compare the updated results with previous ones? Any temporal change?

REVIEWER	Lee, Whanhee Yale University
REVIEW RETURNED	13-Apr-2021

GENERAL COMMENTS	Thank you for giving me a chance to review this wonderful review paper. It was solid and has very high-level completeness. It was difficult to find the flaws of this draft. Nevertheless, I've attached a few comments. Please consider them. Major: 1. 2.5 Data synthesis: The authors used standardized risk estimates. However, it looks difficult to understand the formula.

	Please clarify it or describe it more precisely? In addition, please unify it: standardise/standardize. 2. 2.5. Data synthesis. The authors described that the stratified estimates were pooled in this study. However, I could not find the method for how they were pooled. Was the random effect meta-analysis used? Or, was weighted average used? Please clarify it. 3. 4.5. Strength and limitation. Most of the studies investigated in this review study were from the US or China. Thus, I think there was an inherent and inevitable selection bias, and it also affected the pooled estimates. In other words, there might be a generalizability problem, although there is no proper way to mitigate it. Nonetheless, I hope the authors should consider this issue and address the potential bias as a limitation. Minor: 1. Abstract, Result: "(ii) Impact of short-term exposure ~ was stronger than mortality". In here, does the mortality mean the CVD mortality? Please clarify it.

REVIEWER	Young, S. Stanley CGSTAT
REVIEW RETURNED	27-Apr-2021

GENERAL COMMENTS	61 studies were selected and 41 were used in the meta-analysis. Several points: 1. Figures 2 and 3 are images and the relative risks and confidence limits can not be easily extracted for analysis. The relative risks and their CLs should be provided in an electronic readable format. Also, analysis weights should be given, see their footnote. 2. All the studies used examined multiple outcomes, multiple predictors and multiple covariates. None of the studies did statistical correction for multiple testing or multiple modeling. Numbers coming from these studies are likely biased and so they are unsuitable for meta-analysis. Meta-analysis requires that the numbers used be unbiased estimators of the target value. 3. Figures 2 and 3. I^2 measures study to study heterogeneity. High values argue against there being a common relative risk and argue that important confounding variables might be driving the relative risk rather than the predictors used, BC or EC. Each of the selected papers has other air components, any one of which might be associated with a health effect.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dear professor,
Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.
Have a good day!
Kind regards,
Xiaoling Zhang

Question 1. A few language issues should be noted, for example: "short-term exposure to BC or EC were related" were – was.
Answer: Thank you very much for your comments. We have checked and modified the grammar problems in the manuscript.
Question 2. The authors mentioned the $PM_{2.5}$-adjusted and unadjusted results, I wonder why $PM_{2.5}$ mass cocentration should be adjusted in examining the component of $PM_{2.5}$, can the authors comment on this.
Answer: Thank you very much for your comment and it is significant for us. We performed the analyses which were enlightened by Mostofsky E et al. 2012. We have learned and referred to the research idea introduced in this methodological article. $PM_{2.5}$ may be related to both constituent concentrations and health outcomes. Constituents that are more strongly correlated with $PM_{2.5}$ may appear more closely related to adverse health outcmes than other constituents even if they are not inherently more toxic. So the impact of $PM_{2.5}$ may misjudge the estimate size of the component of $PM_{2.5}$ and it is important to properly account for potential confounding bias by $PM_{2.5}$ in these analyses. Mostofsky E et al. 2012 summarized the advantages and disadvantages of $PM_{2.5}$-adjusted and unadjusted models for the association between particle constituents and health outcomes. When researching the impact of the component of $PM_{2.5}$ to health or diseases, we usually adjust $PM_{2.5}$ and even other component of $PM_{2.5}$, if we focus on just one component. Therefore, we have referred to the methodological article mentioned above and the $PM_{2.5}$-adjusted and unadjusted models have been performed in this manuscript to investigate the robustness of the results.
Question 3. It is interesting to observe that the effects were larger among Asian population, why stronger effects were detected in Asia, more explanations are needed.
Answer: Thank you very much for your comment and it is important for us. We did subgroup analysis to show the association between BC and cardiovascular mortality on the geographical location. And we found significant association between BC and cardiovascular mortality in Asia. We have extracted and calculated the regional distribution of BC concentration of included studies. It showed that the mean BC concentration is 0.984 $\mu g/m^3$ in America, 1.513 $\mu g/m^3$ in Europe and 4.477 $\mu g/m^3$ in Asia, which maybe an essential reason of the results. In addition, we have added it in the discussion. Thank you very much for your comments. So the large effects were investigated among Asia populations maybe caused by high BC concentration in Asia studies which were included in our study. (Page #27)
Question4. A few systematic reviews have been available in the literature, I agree that it is important to update the evidence, the authors, could please, compare the updated results with previous ones? Any temporal change?
Answer: Thank you very much for your comment and it is significant for us. (1) In addition to updating the evidence, we used an adaptation of the GRADE (Grading of Recommendations assessment, Development and Evaluation) framework, formulated by the WHO global air quality guidelines working group to evaluate the certainty of evidence. Because the quality of evidence is essential for guideline and policy development; (2) With comprehensive search databases up to July 19th, 2021. We have included 9 newly published studies in the impact of black carbon on cardiovascular and respiratory diseases; (3) Compared with Yang Y et al. 2019, we performed deeper analyses on the association between black carbon and cardiorespiratory diseases. The results showed that BC was associated with cardiovascular diseases in the/span>elderly and children asthma.
Reference
[1] Mostofsky E, Schwartz J, Coull B A, et al. Modeling the association between particle constituents of air pollution and health outcomes[J]. American journal of epidemiology, 2012, 176(4): 317-326.
[2] Turner M C, Jerrett M, Pope III C A, et al. Long-term ozone exposure and mortality in a large prospective study[J]. American journal of respiratory and critical care medicine, 2016, 193(10): 1134-1142.

[3] Zhang Z, Rowan N R, Pinto J M, et al. Exposure to particulate matter air pollution and anosmia[J]. JAMA Network Open, 2021, 4(5): e2111606-e2111606.
[4] Peng R D, Chang H H, Bell M L, et al. Coarse particulate matter air pollution and hospital admissions for cardiovascular and respiratory diseases among Medicare patients[J]. JAMA, 2008, 299(18): 2172-2179.
[5] Di Q, Dai L, Wang Y, et al. Association of short-term exposure to air pollution with mortality in older adults[J]. JAMA, 2017, 318(24): 2446-2456.
[6] Newbury J B, Arseneault L, Beevers S, et al. Association of air pollution exposure with psychotic experiences during adolescence[J]. JAMA psychiatry, 2019, 76(6): 614-623.
[7] Yang Y, Ruan Z, Wang X, et al. Short-term and long-term exposures to fine particulate matter constituents and health: A systematic review and meta-analysis[J]. Environmental pollution, 2019, 247: 874-882.

Reviewer: 2
Dear editor,
Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.
Have a good day!
Kind regards,
Xiaoling Zhang
Question 1. 2.5 Data synthesis: The authors used standardized risk estimates. However, it looks difficult to understand the formula. Please clarify it or describe it more precisely? In addition, please unify it: standardise/standardize.
Answer: Thank you very much for your comment. We referred to the formula used in the study performed by Shah ASV et al. 2013.

The formula could be explained as following. First, we converted RR into lnRR.
Second, we converted the increment, used standard increment to divide original increment (standard increment here means 1) and got IncreS/IncreO. Third, we used lnRR to multiply IncreS/IncreO and got lnRR*(IncreS/IncreO). Finally, we used the exponential function based on the natural constant "e" and got exp[lnRR*(IncreS/IncreO)]. These are four calculation steps of the formula. For example, RR = 1.023 (per 2.6 μg/m^3). First, lnRR = ln1.023 =
0.023. Second, IncreS/IncreO = Incre1/Incre2.6 = 1/2.6 = 0.385. Third, ln1.023*(1/2.6) = 0.009. Finally, exp[lnRR*(IncreS/IncreO)] = exp[ln1.023*(1/2.6)] =
1.009. We have also unified it as "standardize". Thank you for pointing out this mistakespan style="font-family:'Times New Roman'"> and we have corrected it. (Page #10)
Question 2. 2.5. Data synthesis. The authors described that the stratified estimates were pooled in this study. However, I could not find the method for how they were pooled. Was the random effect meta-analysis used? Or, was weighted average used? Please clarify it.
Answer: Thank you very much for your comment and it is significant for
us. In statistical analysis part, we used the random-effects model for anticipating significant heterogeneity among studies in the meta analysis. Please see details in "2.8. Statistical analysis". (Page #11)
Question 3. 4.5. Strength and limitation. Most of the studies investigated in this review study were from the US or China. Thus, I think there was an inherent and inevitable selection bias, and it also affected the pooled estimates. In other words, there might be a generalizability problem, although there is no proper way to mitigate it. Nonetheless, I hope the authors should consider this issue and address the potential bias as a limitation.
Answer: Thank you very much for your comment and it is significant for us. Your point of view is essential and reasonable. And we have added this selection bias in the potential bias as a limitation. (Page #27)

Question 4. Abstract, Result: "(ii) Impact of short-term exposure ~ was stronger than mortality". In here, does the mortality mean the CVD mortality? Please clarify it.
Answer: Thank you very much for your comment and it is important for us. Yes, here the mortality means the CVD mortality. We have clarified it in "Abstract, Result". (Page #3)
Reference
[1] Mostofsky E, Schwartz J, Coull B A, et al. Modeling the association between particle constituents of air pollution and health outcomes[J]. American journal of epidemiology, 2012, 176(4): 317-326.
[2] Shah A S V, Langrish J P, Nair H, et al. Global association of air pollution and heart failure: a systematic review and meta-analysis[J]. The Lancet, 2013, 382(9897): 1039-1048.

Reviewer: 3
Dear editor,
Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.
Have a good day!
Kind regards,
Xiaoling Zhang
Question 1. Figures 2 and 3 are images and the relative risks and confidence limits cannot be easily extracted for analysis. The relative risks and their CLs should be provided in an electronic readable format. Also, analysis weights should be given, see their footnote.
Answer: Thank you very much for your comment and it is significant for us. We have provided an electronic readable format of estimates and confidence intervals. And the analysis weights have been given in the footnote. Please check Supplementary material (Reviewer 3) and we have listed all the data. [Supplementary material (Reviewer 3)]
Question 2. All the studies used examined multiple outcomes, multiple predictors and multiple covariates. None of the studies did statistical correction for multiple testing or multiple modeling. Numbers coming from these studies are likely biased and so they are unsuitable for meta-analysis. Meta-analysis requires that the numbers used be unbiased estimators of the target value.
Answer: Thank you very much for your comment and it is significant for us. (1) Meta-analysis of observational studies has limitations with inherent biases. Therefore, we performed the meta-analysis with repeated verification and comprehensive consideration. We did $PM_{2.5}$ adjusted and unadjusted model analysis, sensitive analysis on all age populations by the removal of studies with partial temporal overlap from the same geographical location to investigate the robustness of results. So caution should therefore be exercised when interpreting the pooled risk estimates and highlights the need for more high-quality data to increase the certainty of these risk estimates; (2) Although the data came from the included studies may have biases, it is still significant for current research progress. Evidence is also important in formulating clinical practice guidelines which could provide scientific basis to guidelines; (3) We referred to some high-quality researches and used some methods to reduce the confounding bias. We chose the random-effects model in our study and it is suitable for anticipated significant heterogeneity between studies in terms of population and methods in this condition. Estimates from the maximally adjusted model in the cohort study were extracted when multiple estimates were present in the original study to reduce the risk of potential unmeasured confounding bias; (4) We pooled relative risks for a standardized increment in pollutant concentration and calculated standardized risk estimates for each study using formula. And we also did subgroup analysis to reduce the confounding bias.
Question 3. Figures 2 and 3. I^2 measures study to study heterogeneity. High values argue against there being a common relative risk and argue that important confounding variables might be driving the relative risk rather than the predictors used, BC or EC. Each of the selected papers has other air components, any one of which might be associated with a health effect.

Answer: Thank you very much for your comment and it is significant for us. (1) Air pollution is a kind of risk factor which may affect human's health. It has a large number of components and makes it complex. So when we focus on only one component, it is difficult to separate other components and they inevitably mixed up together. But there are some methods to minimize the confounding bias and give prominence to the effects of target pollutants; (2) We chose the random-effects model in our study and it is suitable for anticipated significant heterogeneity between studies in terms of populations and methods in this condition; (3) We pooled relative risks for a standardized increment in pollutant concentration and calculated standardized risk estimates for each study using formula. And we also did subgroup analysis to reduce the confounding bias; (4) It is true that the estimates should not be pooled when I^2 is high, but the results which came from more rigorous research methods have guiding significance for the present research progress. Evidence is also important in formulating clinical practice guidelines which could provide scientific basis to guidelines. As for the research field in the association between air pollution and health, meta-analysis of observational studies has limitations with inherent biases. We referred to some high-quality researches and used some methods to reduce the confounding bias. $PM_{2.5}$-adjusted estimates and $PM_{2.5}$-unadjusted estimates were combined respectively to investigate the robustness of results and we found they are robust. This method came from Mostofsky E et al. 2012, a methodological article. They summarized the advantages and disadvantages of $PM_{2.5}$-adjusted and unadjusted models for the association between particle constituents and health outcomes. For example, Yang B Y et al. 2019 regressed highly correlated air pollutants against each other and used model residuals as covariates to control for potential confounding bias by coexposure to other pollutants. $PM_{2.5}$ has also been adjusted to examine the impact of $PM_{10-2.5}$ in Peng R D et al. 2008, they tried to minimize the effect of $PM_{2.5}$. Turner M C et al. 2016 have also adjusted $PM_{2.5}$ and NO_2 to investigate the association between O_3andcardiorespiratory mortality. $PM_{2.5}$ (adjusted by O_3) and warm-season O_3 (adjusted by $PM_{2.5}$) were estimaed to investigate the association with all-cause mortality in Di Q et al. 2017.

VERSION 2 – REVIEW

REVIEWER	Lee, Whanhee Yale University
REVIEW RETURNED	24-Aug-2021

GENERAL COMMENTS	I think this paper is ready for the publication.

REVIEWER	Young, S. Stanley CGSTAT
REVIEW RETURNED	14-Sep-2021

GENERAL COMMENTS	Referee 3 reply bmjopen-2021-049516 Analysis based on **File3 Supplementary material (Reviewer 3).docx** Tables 1 and 2 The authors provided computer readable risk ratios and CLs for cardiovascular and respiratory effects, followed by Referee3 comments on revision **Analysis Method** We use a relatively old plotting technique, a p-value plot, to examine multiple tests visually. First, an example to get a sense of the method. The story is simple and rather funny. Three researchers claimed that eating breakfast cereal would cause more male babies. They had run a survey, Food Frequency Questionnaire, FFQ. In an FFQ, subjects are asked to recall the foods they ate using a standard questionnaire. FFQs started with 61 foods. The number of foods in this FFQ was 131 at two time points. Sometime later, in this case 9 months later, a health inventory is taken. The number of health effects can be 20 or more. Analysis is mix

and match. This presents a multiple testing problem that is essentially never considered by FFQ researchers. Here is the title of their paper, “You are what your mother eats: evidence for maternal preconception diet influencing foetal sex in humans” which, on the face of it, is wildly unreasonable. Genetics would say X/Y from the father is determinative. Our reply title was “Cereal-induced gender selection? Most likely a multiple testing false positive”.
In this case 131 foods are included in the FFQ at two time points. A p-value plot, ranked p-values plotted against the integers, shows that the results are consistent with randomness.
Figure 1. P-value plot. Ranked p-values are plotted against the integers.

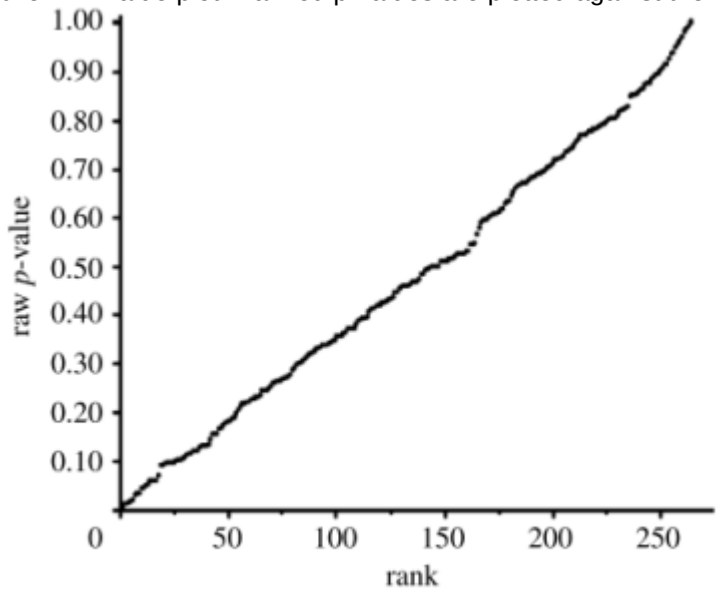
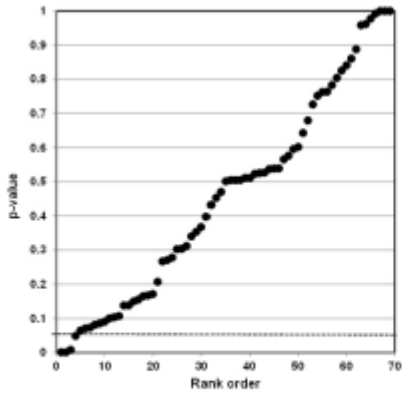
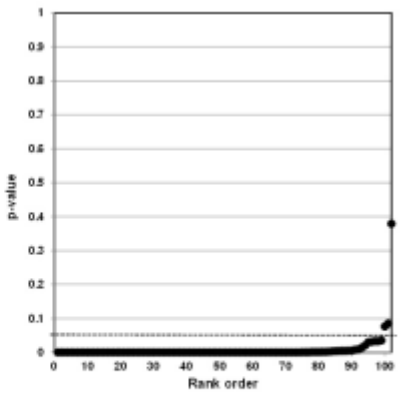


Figure 1. The *p*-value plot of 262 *p*-values.

What is expected behavior of a p-value plot?
Figure 2. Left: null hypothesis, 45-degree line. Right: most p-values below 0.05.



(i) elderly long-term exercise training–mortality & morbidity risk (n=69)



(ii) smoking–squamous cell carcinoma risk (n=102)

Both plots conform to distinct (single) sample distributions – for null and real effects. Any p-value that does not conform to one of these behaviors should be regarded as suspect.
One source of p-values examining a single scientific question is a meta-analysis. Meta-analysis is a statistical analysis that combines the results of multiple scientific studies: (1) It involves searching literature to identify statistically similar studies and then further screening & reviewing to narrow down a list of studies whose data are combined & analyzed (quantitative analysis). (2) It is intended to address a single research question… e.g., does *factor A* cause *disease B*? (3) Meta-analysis

techniques are highly regarded in medical sciences. (4). However, meta-analysis of poorly designed studies produces erroneous statistics and may be misleading. Construction of a p-value plot for a meta-analysis proceeds as follows.

1. Risk ratios and Confidence Limits are used to compute a p-value for each study.
2. P-Values are ranked from smallest to largest and plotted against the integers 1, 2, 3, ….

If the p-value plot is bilinear, has the shape of a hockey stick, the results are ambiguous. The small p-values on the blade support a real effect. The p-values on the handle support no effect. The small p-values might be correct (real), or they could arise from p-hacking or even from fraudulent data. In any case the results are ambiguous. We have examined over 50 air quality papers/health effect papers. The median number of possible analyses is on the order of 10,000 indicating there is an opportunity for p-hacking. We have asked for over 100 air quality/health effects data sets. We have been given only a few.

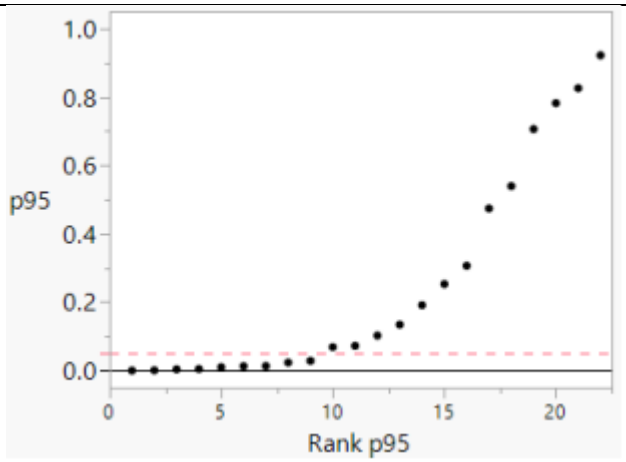
Comment: The random effects analysis, DerSimonian and Laird 1986, is a weighted average, and it is presumed that all the studies have the same underlying process going on. Keep in mind that an average is not a fair summary of data if there are outliers, e.g. some of the studies are p-hacked or use fraudulent data.

Analysis Song 2

Analysis of Table 1, cardiovascular

RowID	AEU	Author	Year	RR	CIlow	CIhigh	SE95	Z95	p95	Rank p95
1	2	Ostro	2015a	0.994	0.953	1.038	0.0217	-0.2767	0.782006	20
2	2	Ostro	2015a	1.005	0.979	1.031	0.0133	0.3769	0.706231	19
3	2	Atkinson	2016	0.987	0.973	1.001	0.0071	-1.8200	0.068759	10
4	1	Geng	2013	1.012	1.002	1.021	0.0048	2.4758	0.013294	7
5	0	Liu	2016a	0.960	0.857	1.076	0.0559	-0.7160	0.474003	17
6	0	Liu	2016b	1.020	0.858	1.214	0.0908	0.2202	0.825696	21
7	0	Sarnat	2015	1.038	1.005	1.073	0.0173	2.1906	0.028482	9
8	0	Kim	2012	1.056	1.018	1.094	0.0194	2.8884	0.003872	3
9	1	Wang	2019a	1.011	0.999	1.023	0.0061	1.7967	0.072389	11
10	3	Maynard	2007	1.076	0.980	1.179	0.0508	1.4971	0.134371	13
11	0	Winquist	2015	1.048	1.012	1.085	0.0186	2.5775	0.009951	5
12	0	Tolbert	2007	1.013	1.004	1.022	0.0046	2.8311	0.004639	4
13	1	Gong	2019	1.002	1.001	1.003	0.0005	3.9200	0.000089	1
14	3	Ostro	2007	1.026	1.004	1.049	0.0115	2.2649	0.023519	8
15	0	Metzger	2004	1.017	1.007	1.027	0.0051	3.3320	0.000862	2
16	3	Kim	2015	1.031	0.935	1.133	0.0505	0.6137	0.539389	18
17	1	Huang	2012	1.005	0.998	1.010	0.0031	1.6333	0.102399	12
18	1	Son	2012	1.001	0.981	1.021	0.0102	0.0980	0.921932	22
19	1	Heo	2014	1.006	0.994	1.017	0.0059	1.0226	0.306493	16
20	2	Basagana	2015	0.979	0.944	1.016	0.0184	-1.1433	0.252900	15
21	0	Basagana	2015	1.026	1.006	1.047	0.0105	2.4859	0.012924	6
22	1	Lin	2016a	1.002	0.999	1.005	0.0015	1.3067	0.191326	14

The study p-values, p95, range from 0.000089 to 0.921932. There are nine p-values less than 0.05. Thirteen p-values >0.05 form a roughly 45-degree line. Boos and Stefanski, 2011 note that if experimental technique is good that an initial p-value <0.001 should essentially always replicate. Certainly, you should not see 13/22 non-replications. At best the results are ambiguous, at worst there is significant p-hacking and possible outright data fraud.

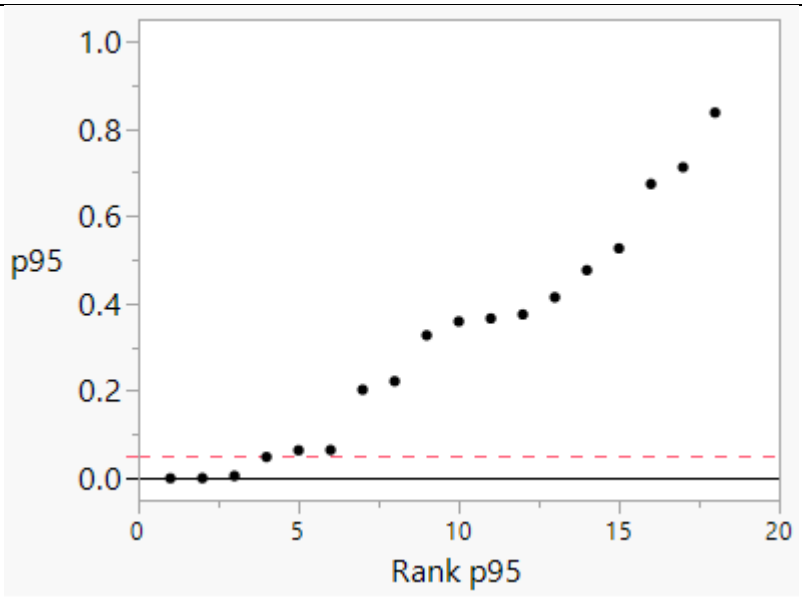


The p-value plot is bilinear, has the shape of a hockey stick.
Boos DD, Stefanski LA. 2011. P-value precision and reproducibility. The American Statistician. 65:213−221. doi:10.1198/tas.2011.10129.

Analysis of Table 2, respiratory

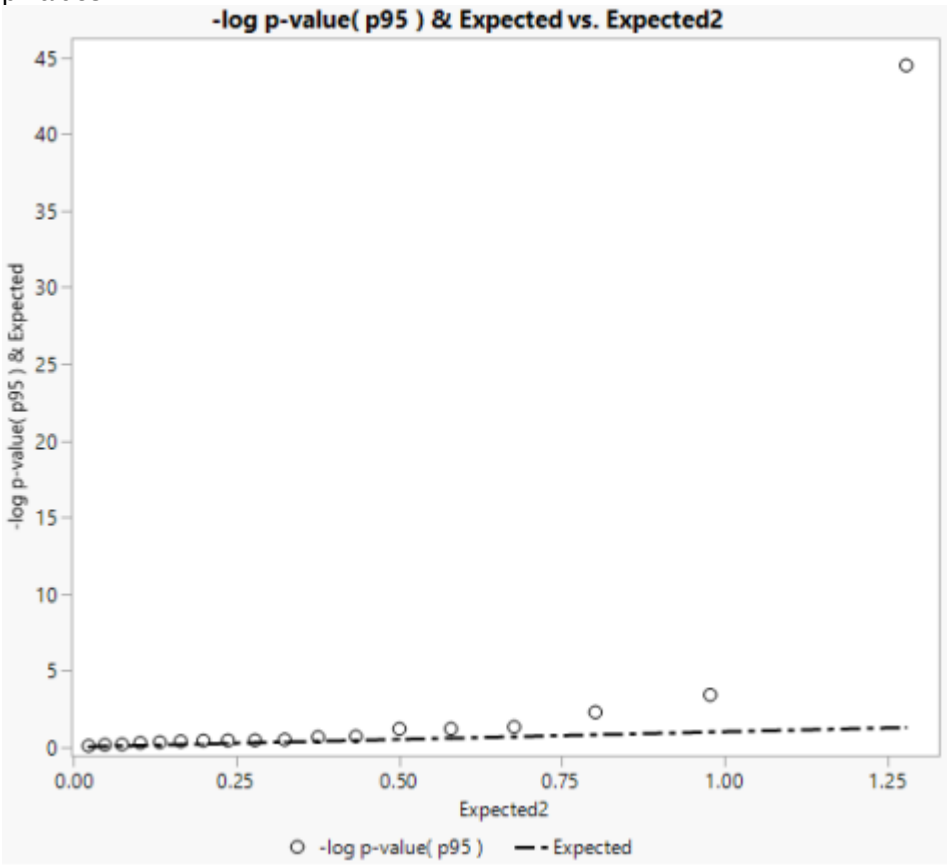
RowID	Author	Year	RR	CLlow	CLhigh	SE95	Z95	p95	Rank p9
1	Atkinson	2016	1.013	0.993	1.033	0.0102	1.2740	0.20266342	
2	Geng	2013	1.002	0.983	1.021	0.0097	0.2063	0.83654425	1
3	Ostro	2015a	1.090	1.004	1.183	0.0457	1.9709	0.04872963	
4	Ostro	2015a	1.064	1.020	1.110	0.0230	2.7876	0.00531073	
5	Sarnat	2015	0.995	0.969	1.022	0.0135	-0.3698	0.71152308	1
6	Huang	2012	1.005	0.993	1.017	0.0061	0.8167	0.41411895	1
7	Son	2012	0.989	0.956	1.024	0.0173	-0.6341	0.52600405	1
8	Kim	2015	1.081	0.920	1.266	0.0883	0.9177	0.3587823	1
9	Heo	2014	0.988	0.962	1.015	0.0135	-0.8875	0.37478438	1
10	Basagana	2015	0.986	0.949	1.026	0.0196	-0.7127	0.47601454	1
11	Basagana	2015	0.940	0.879	1.006	0.0324	-1.8520	0.06403035	
12	Maynard	2007	1.196	1.005	1.421	0.1061	1.8469	0.06475829	
13	Liu	2016a	0.964	0.895	1.039	0.0367	-0.9800	0.32708612	
14	Liu	2016b	0.963	0.806	1.150	0.0878	-0.4216	0.67329663	1
15	Kim	2012	1.100	0.949	1.270	0.0819	1.2212	0.22201644	
16	Cakmak	2009	1.036	1.031	1.041	0.0026	14.1120	3.2036e-45	
17	Wang	2019a	1.038	1.017	1.059	0.0107	3.5467	0.00039014	
18	Tolbert	2007	0.997	0.990	1.003	0.0033	-0.9046	0.36566918	1

The smallest p-value is 3.2×10^{-45}. The largest p-value is .8365. The small p-value is so small as to indicate possible data fraud. Four p-values are less than 0.05 while fourteen are larger than 0.05 and fall on an approximate 45-degree line.



The p-value plot is bilinear, Hockey Stick appearance. At best the results are ambiguous; at worst there is possible/likely data fraud.

Alternative p-value plot, respiratory.
Here we plot the negative log base 10 against the expect values of ranked uniform p-values.



We see that most of the p-values are close to their expectations, indicating no effect in most of the studies. Three dramatically exceed their expectations. One p-value is remarkably small (large -Log10). This p-value is suspect (possible data fraud).

Analysis conclusions

	Both cardiovascular and respiratory p-value plots give ambiguous results: Some p-values are very small while many p-values fall on an approximate 45-degree line indicating random results. This analysis comes down on the side there is no evidence for a consistent effect. There is evidence for p-hacking or data fraud. Knowing the data is not consistent with a health effect is worthwhile knowledge. I support a re-write of the paper along these lines. I'm willing to break the blind and help (provide additional references and compute analyses) if that is the direction the editor wants.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 3

Dear reviewer 3,

Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.

Have a good day!

Kind regards,

Xiaoling Zhang, Shigong Wang

Comments: After analysis of the provided data, I think the study is negative. There is not strong evidence in support of a positive claim and there is evidence that some of the base paper obtained small p-values by p-hacking or by use of fraudulent data.

If you choose to go in that direction I will help with literature and analysis.

Answer: Thank you very much for your professional comments. The reliability of air quality epidemiological studies is often poor, with a serious lack of reproducibility of published findings. If researchers run a regression with and without outliers, with and without a covariate, with one and then another dependent variable, and false positive results are much more likely to be reported. It makes no sense to do meta-analysis and likely obtain positive results, without considering the reliability of the p-values, possible p-hacking. Therefore, it is essential to explore the p-values used in a meta-analysis. We have added the analyses of search space counts, p-value calculation and p-value plot in the manuscript. In addition, we also rewrite the corresponding content. We have learned a lot from your comments.

VERSION 3 – REVIEW

REVIEWER	Young, S. Stanley CGSTAT
REVIEW RETURNED	01-Mar-2022

GENERAL COMMENTS	This paper represents considerable work. The systematic review appears sound. The meta-analysis calculations were done with a random effects model. GRADE was used to evaluate the reliability of the base studies. It is important to note the there is no provision within GRADE for evaluating multiple testing or multiple modeling.

	The authors count out a random sample of papers and discover that the analysis search space is typically large for a base paper. The median number of possible analyses is 12,000. It is the obligation of a researcher making a claim that strong supporting evidence be provided. The base papers used in their meta-analysis do not correct for multiple testing and multiple modeling, MTMM, and therefore do not protect from MTMM bias. Authors of base papers do not provide strong evidence for their claims. Also, a p-value plot shows ambivalent results – some studies support a positive effect, while others appear random. The authors present meta-analysis results, counting results and p-value plots allowing the reader to form an opinion of the results.

VERSION 3 – AUTHOR RESPONSE

Reviewer: 3
Dear reviewer 3,
Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.
Have a good day!
Kind regards,
Xiaoling Zhang, Shigong Wang
Comments: This paper represents considerable work. The systematic review appears sound. The meta-analysis calculations were done with a random effects model. GRADE was used to evaluate the reliability of the base studies. It is important to note the there is no provision within GRADE for evaluating multiple testing or multiple modeling. The authors count out a random sample of papers and discover that the analysis search space is typically large for a base paper. The median number of possible analyses is 12,000.
It is the obligation of a researcher making a claim that strong supporting evidence be provided. The base papers used in their meta-analysis do not correct for multiple testing and multiple modeling, MTMM, and therefore do not protect from MTMM bias. Authors of base papers do not provide strong evidence for their claims. Also, a p-value plot shows ambivalent results – some studies support a positive effect, while others appear random.
The authors present meta-analysis results, counting results and p-value plots allowing the reader to form an opinion of the results. [Please see attached file for additional comments]
Answer: Thank you very much for your professional comments. We have added the analyses of search space counts, p-value calculation and p-value plots in the manuscript. In addition, we have rewritten the corresponding content and revised the problems you mentioned.

VERSION 4 – REVIEW

REVIEWER	Young, S. Stanley CGSTAT
REVIEW RETURNED	23-Mar-2022

GENERAL COMMENTS	Small edits and clarifications: Page 7 line 19 “but p-hacking is a common issue” Line 35 p-hacking Line 43 in a meta-analysis Page 8 Line 12

	was performed examined Page 11 Line 51 Evaluation of reliability of evidence There are multiple uses of the term “certainty of evidence” . That may be standard terminology. I prefer the term reliability. Page 15 line14 The number of… Page 15, line 32 “If the shape of p-value plot is a straight line, the p-values are in a distribution of true null hypothesis.31 If the shape follows an approximate 45-degree line, the p-values are assumed to be random. If the shape is approximately a hockey stick, the p-values on the blade are unlikely due to chance. Therefore, p-value plot was used to assess the validity and reliability of included studies.” becomes If the shape of p-value plot is a straight line and follows an approximate 45-degree line, then the p-values are consistent with a distribution of true null hypothesis; the p-values are assumed to be random.31 If the shape is approximately a hockey stick, the p-values on the blade are not consistent with chance, whereas those on the arm are consistent with chance, the results are ambiguous. Therefore, p-value plot was used to assess the validity and reliability of included studies. Page 26, line 17 “Mostofsky et al. proved showed that PM2.5 may be” Page 26 line 20 Constituents Page 27, line 53 “Reliability of meta-analysis is an essential challenge existed in environmental epidemiology researches, which should be improved in the future.” The reliability of meta-analysis is an essential challenge for environmental epidemiology research, which should be improved in the future.

VERSION 4 – AUTHOR RESPONSE

Reviewer 3
Dear reviewer 3,
Thank you very much for your valuable suggestions on our paper. We have carefully modified the manuscript according to your comments. We hope you will be satisfied with our revision. Please contact us if you have any other questions.
Have a good day!
Kind regards,
Xiaoling Zhang, Shigong Wang
Comments to the Author:
Small edits and clarifications:
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Line 35

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Page 26 line 20
Constituents

Page 27, line 53
“Reliability of meta-analysis is an essential challenge existed in environmental epidemiology researches, which should be improved in the future.”
The reliability of meta-analysis is an essential challenge for environmental epidemiology research, which should be improved in the future.
Answer: Thank you very much for your professional comments. Each of the modifications you mentioned has been corrected in the manuscript, except for one modification suggestion about "certainty of evidence". We referred to related researches which used GRADE approach or methodological articles. They used "certainty of evidence" or "quality of evidence" in their studies. So we did not revise this suggestion in the manuscript.
Reference：
[1] Jarosinska D, Velasco R P. Approach to assessing the certainty of evidence in environment and health–the case of WHO global air quality guidelines[C]//ISEE Conference Abstracts. 2021, 2021(1).
[2] Arumugam A, Björklund M, Mikko S, et al. Effects of neuromuscular training on knee proprioception in individuals with anterior cruciate ligament injury: a systematic review and GRADE evidence synthesis[J]. BMJ open, 2021, 11(5): e049226.
[3] Morgan R L, Thayer K A, Bero L, et al. GRADE: Assessing the quality of evidence in environmental and occupational health[J]. Environment international, 2016, 92: 611-616.